

Math 526, S06
Quiz 5

Name:
SSN: xxx-xx-

Instructions: There are totally 1 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 2 & 1 & & \\ 1 & 2 & 1 & \\ & 1 & 2 & 1 \\ & & 1 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 3 \\ 5 \end{bmatrix}$.

a) Is $\mathbf{A}$ column diagonally dominant?

Solution: Yes.

b) Find the $\mathbf{L} - \mathbf{U}$ factorization of the tridiagonal matrix $\mathbf{A}$.

Solution:

$$\mathbf{L} = \begin{bmatrix} 1 & & & \\ \frac{1}{2} & 1 & & \\ & \frac{2}{3} & 1 & \\ & & \frac{3}{4} & 1 \end{bmatrix}$$

and

$$\mathbf{U} = \begin{bmatrix} 2 & 1 & & \\ & \frac{3}{2} & 1 & \\ & & \frac{4}{3} & 1 \\ & & & \frac{5}{4} \end{bmatrix}$$

c) Solve the tridiagonal system $\mathbf{Ax} = \mathbf{b}$ using b).

Solution: First we solve $\mathbf{Ly} = \mathbf{b}$, i.e.,

$$\begin{bmatrix} 1 & & & \\ \frac{1}{2} & 1 & & \\ & \frac{2}{3} & 1 & \\ & & \frac{3}{4} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \\ 5 \end{bmatrix}$$

We get

$$y_1 = 1, \quad y_2 = 0 - \frac{1}{2}y_1 = -\frac{1}{2}, \quad y_3 = 3 - \frac{2}{3}y_2 = \frac{10}{3}, \quad y_4 = 5 - \frac{3}{4}y_3 = \frac{5}{2}$$

Then we solve $\mathbf{Ux} = \mathbf{y}$, i.e.,

$$\begin{bmatrix} 2 & 1 & & \\ & \frac{3}{2} & 1 & \\ & & \frac{4}{3} & 1 \\ & & & \frac{5}{4} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ -1/2 \\ 10/3 \\ 5/2 \end{bmatrix}$$

We get

$$x_4 = \frac{5}{2} / \frac{5}{4} = 2, \quad x_3 = \frac{3}{4} \left(\frac{10}{3} - x_4 \right) = 1, \quad x_2 = \frac{2}{3} \left(-\frac{1}{2} - x_3 \right) = -1, \quad x_1 = \frac{2}{3} (1 - x_2) = 1$$

So the solution is

$$\mathbf{x} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 2 \end{bmatrix}$$